

Amazon Sales Data Dashboard

Exploring products, pricing, discounts, and ratings

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CS-470 — Introduction to Artificial Intelligence | Final Project

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THE QUESTION

What can Amazon's sales data tell everyday shoppers?

- 1 Which product categories dominate the marketplace?
- 2 Which products offer the biggest discounts?
- 3 How do actual price and discounted price compare?
- 4 Do higher-rated products cost more?

GOAL

Build an interactive dashboard that lets anyone — technical or not — quickly understand product pricing, discount patterns, and customer ratings across categories.

THE DATASET

Amazon Sales Dataset

Sourced from Kaggle — real Amazon product listings, mostly electronics and home goods

1,465

Products (raw)

16

Fields per product

9

Top-level categories

KEY FIELDS

product_name • category • actual_price • discounted_price • discount_percentage • rating • rating_count

The Data Was Messier Than It Looked

Prices stored as text	<i>"₹1,099" instead of 1099</i>	Stripped the ₹ symbol and commas, converted to numbers
Discounts stored as text	<i>"64%" instead of 64</i>	Stripped the % sign, converted to numbers
Invalid rating values	<i>A placeholder " " instead of a number</i>	Converted safely; invalid values dropped instead of crashing
Review counts with commas	<i>"24,269" plus missing entries</i>	Stripped commas, dropped rows still missing

Result: 1,465 rows in → 1,462 usable rows after cleaning (only 3 dropped for missing data)

At a Glance

Summary Metrics

Total Products

1,462

Average Rating

4.10 / 5

Average Discount

48%

Average Actual Price

₹5,453

Average Discounted Price

₹3,130

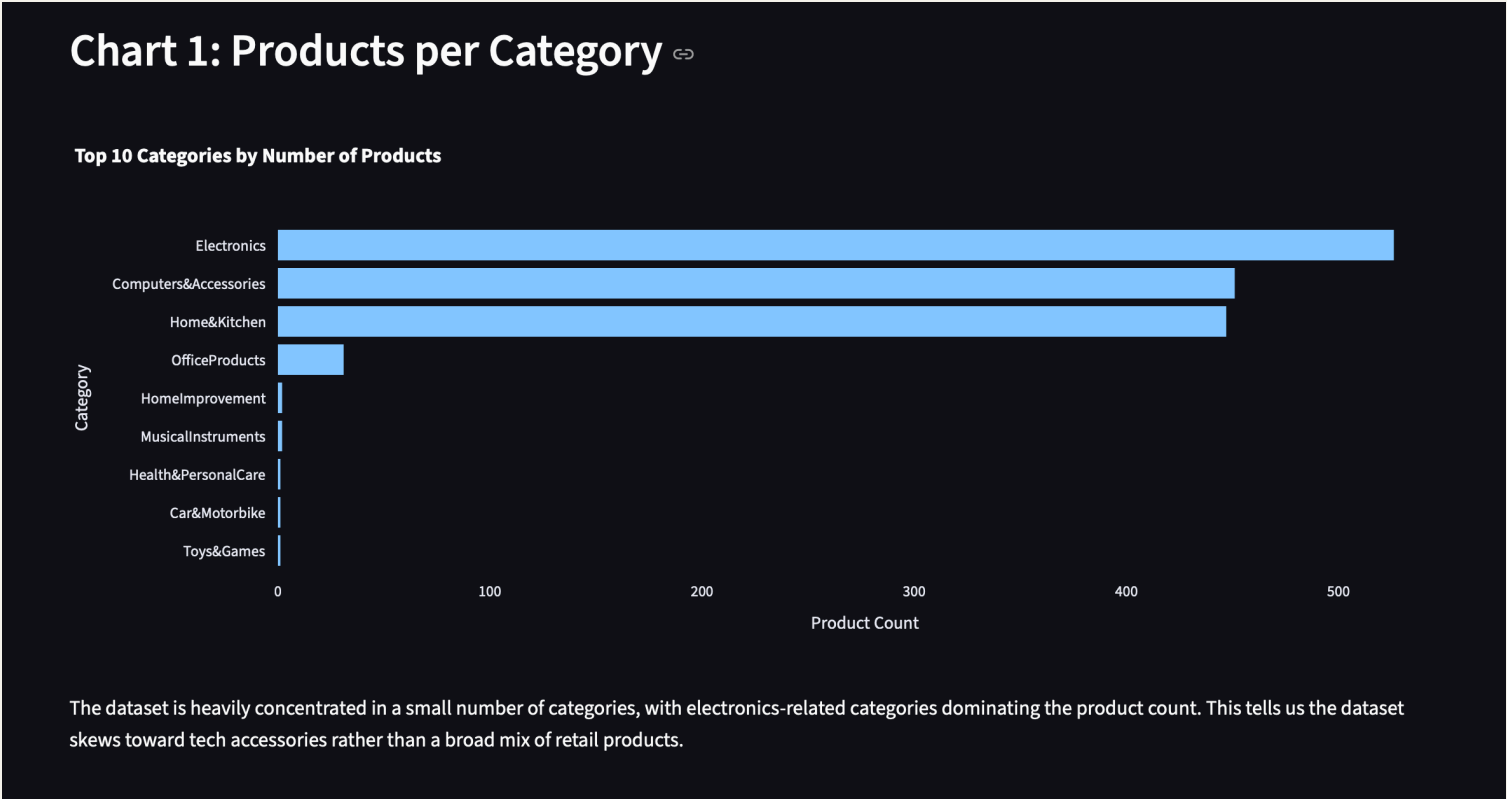
Categories Represented

9

1,462 products across 9 categories, discounted an average of 48% off an average listed price of ₹5,453.

FINDING 1

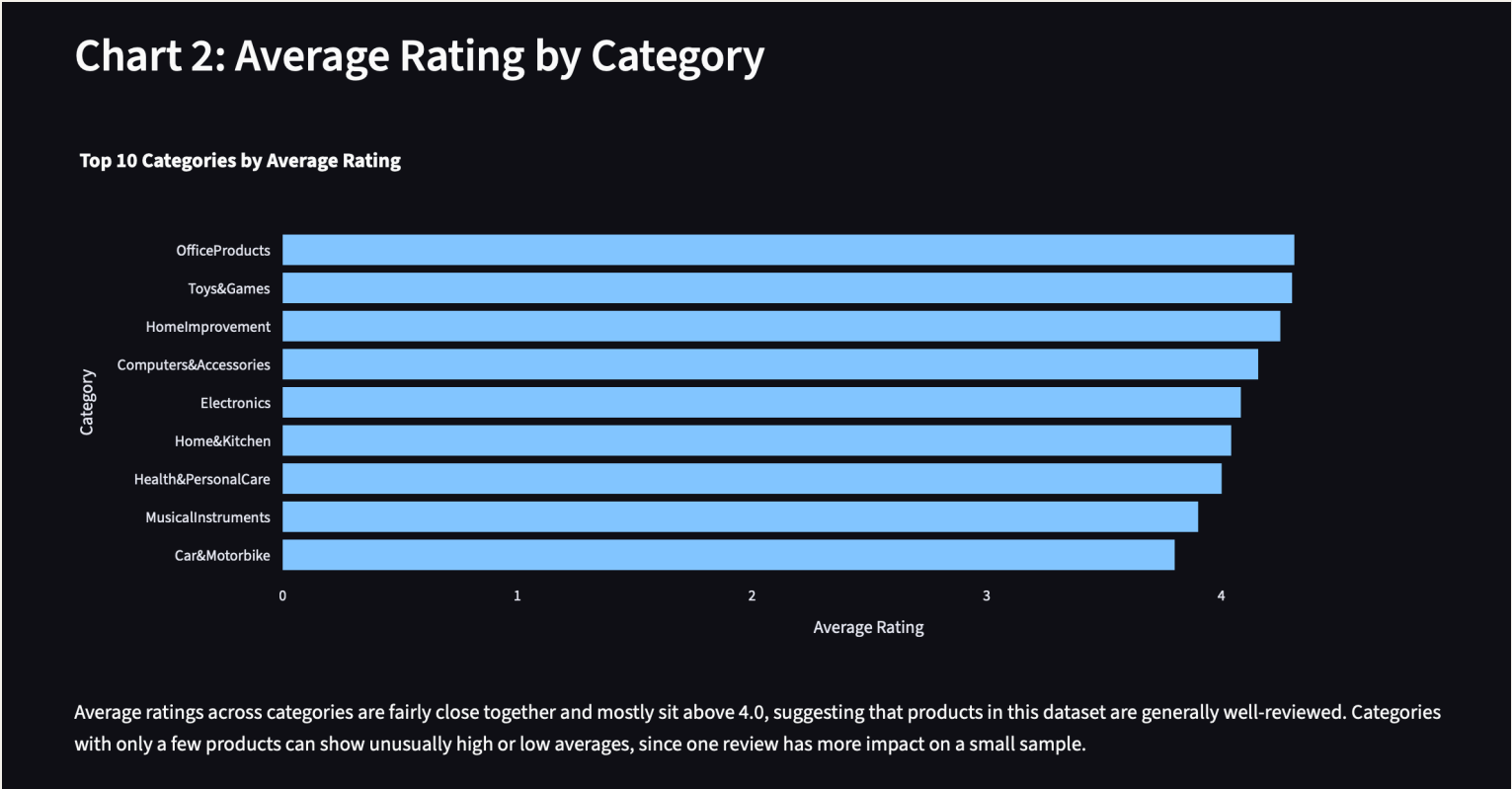
Which Categories Have the Most Products?



The dataset is heavily concentrated in a small number of categories, with electronics-related categories dominating the product count — telling us this dataset skews toward tech accessories rather than a broad mix of retail products.

FINDING 2

Which Categories Rate Highest?



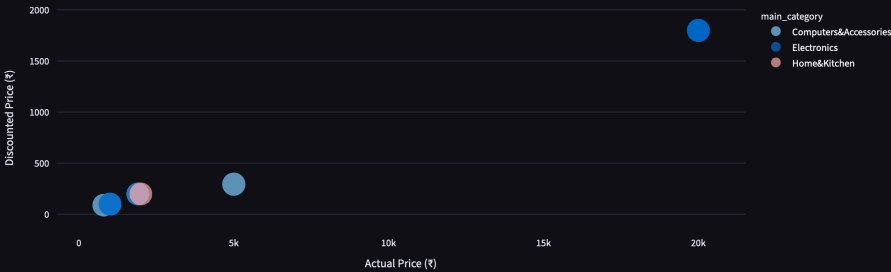
Average ratings across categories are fairly close together and mostly sit above 4.0, suggesting products in this dataset are generally well-reviewed. Categories with only a few products can show unusually high or low averages.

FINDING 3

Price and Rating Patterns

Chart 3: Actual Price vs. Discounted Price

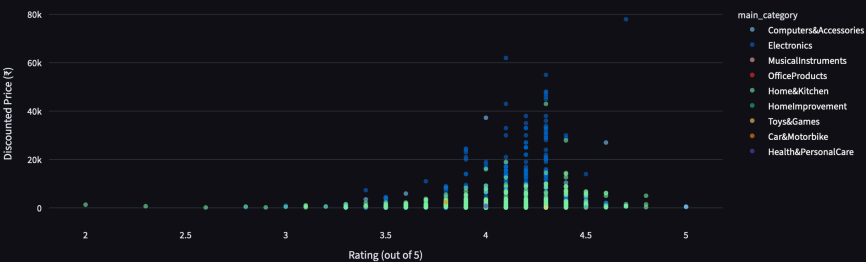
Top 15 Products by Discount Percentage: Actual vs. Discounted Price



Among the most heavily discounted products, the gap between actual and discounted price is large, with some items discounted by well over 50%. Higher original prices don't guarantee a bigger discount, so shoppers should compare discount percentage directly rather than assuming expensive items are marked down the most.

Chart 4: Rating vs. Discounted Price

Product Rating vs. Discounted Price



There is no strong relationship between rating and price in this dataset — both cheap and expensive products can be rated highly. This suggests customers are rating based on product quality and value rather than simply rating expensive items higher.

Data source: Amazon Sales Dataset (Kaggle) — amazon.csv

Among the most-discounted products, the gap between actual and discounted price is large — some items are marked down over 50%, though a higher original price doesn't guarantee a bigger discount. And rating shows no strong link to price: both cheap and expensive products can be rated highly.

PROCESS

How I Used AI Tools

- **Planning**
Asked Claude to read the assignment and outline the build steps in order
- **Data Cleaning**
Asked for code to handle ₹ symbols, %, commas, and the bad rating value — then verified the row count myself
- **Dashboard Build**
Asked Claude to build the Streamlit app structure, then ran and checked every chart against the raw data
- **Debugging**
Got help fixing uv install, PATH, and .zshrc permission issues — made my own call not to use sudo

Conclusions & Next Steps

WHAT WE LEARNED

- The dataset skews heavily toward electronics and tech accessories
- Most products are well-reviewed, averaging 4.10 out of 5
- Discounts average 48%, but a higher original price doesn't guarantee a bigger markdown
- Rating and price show no strong relationship — quality isn't just about cost

WHAT'S NEXT

- Bring in review text for sentiment analysis alongside star ratings
- Track price history over time instead of a single snapshot
- Compare this dataset against other regions or marketplaces
- Let users filter the dashboard by category or price range

Thank you!